

Melbourne Airbnb Review Activity

Interim Project Report

CMCE30005 Business Analytics Challenge | Semester 2 2026 | TheNextChapter Group 2

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Repository: [Qihang-cpu/CMCE30005-TheNextChapter](https://github.com/Qihang-cpu/CMCE30005-TheNextChapter)

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Introduction

A prospective operator considering several Melbourne Airbnb properties must compare locations and dwelling configurations before committing to leases, furnishings and launch costs. These choices are difficult to reverse and an unsuitable property can tie up limited capital. In this report, location is measured by Local Government Area (LGA), meaning a municipal area within Greater Melbourne. The business therefore needs a consistent way to compare similar properties rather than relying on nightly price alone.

The school-supplied Inside Airbnb snapshot shows substantial variation in quoted prices and guest reviews. We combine descriptive comparisons with predictive modelling to assess which market segments are associated with stronger review activity. Reviews are observable and provide a limited indicator of guest activity, but they are not bookings or profit. Because the supplied data contain neither actual rent nor operating costs, this interim analysis supports preliminary market screening rather than a financial investment recommendation.

Problem Definition and Objectives

Our research question is: Among established standard entire-home listings with one to three bedrooms in Greater Melbourne, which LGA, dwelling-class and bedroom-count segments with at least 50 eligible analysis listings have the highest mean out-of-sample predicted probability of meeting a common upper-quartile review-count benchmark over the 365 days ending on each listing's scrape date?

In practical terms, we compare similar Melbourne property segments to identify where strong recent review activity is most common. Descriptive analysis compares review distributions, benchmark attainment and property attributes. Predictive analysis tests whether location, dwelling configuration, capacity, bathrooms and amenities distinguish higher-activity listings when the model is evaluated on different hosts from those used for training.

Established means the first observed review occurred at least 365 days before that listing's scrape date. This indicates review history, not an opening date or continuous operation. We retain entire rental units, condos, homes and townhouses with one to three bedrooms and quoted prices of AUD30–1,500. Units and condos form the apartment class; homes and townhouses form the house class. These categories identify comparable residential stock without proving lease or subletting availability. Each reported segment needs at least 50 analysis listings after all restrictions.

A separate group of reference hosts supplies one common upper-quartile benchmark. Its numerical value is calculated from the supplied data, rather than fixed in the research question. It is neither a whole-market standard nor a profitability threshold.

Data Description

The school's June 2026 files contain 25,728 listings, 9,390,720 calendar records and 1,026,690 reviews. The 90 listing variables cover identifiers, LGA, property type, capacity, bathrooms, amenities, price, minimum stay and reviews. Scrape dates range from 17 June to 1 July 2026. Calendar availability cannot establish bookings and the calendar contains no prices.

Cleaning converts currency strings, standardises missing values, extracts bathroom counts and parses amenity lists without splitting names containing commas. Quoted price is missing for 6,553 listings and bedrooms for 4,679. Median price is AUD243.67 across 19,175 non-missing records. The price restriction supports comparability; it does not establish that excluded prices are errors. Source identifier text is preserved because scientific notation can already contain irrecoverable rounding.

The final reference group's P75 is 29.75, rounded upward to a 30-review event. The separate analysis group contains 3,873 listings from 1,726 hosts across 14 segments, including 334 zero-review listings. Of these, 981 meet the benchmark (25.33%); ties are retained. Median amenity counts are 44 among listings meeting the benchmark and 43 below it, while median guest capacity is four in both groups. These pooled differences offer limited separation.

Full-source checks revealed a one-day boundary difference: the supplied recent-review field matches the inclusive interval from scrape date minus 365 days through scrape date, covering 366 dates. We retain that field and derive reviews_365d over the strict 365-day interval, excluding its start boundary. This removes 352 reviews across 350 listings and changes one analysis listing's target label. Independent reconstruction agrees with every derived count; original total-review counts and first/last review dates also match.

Table 1. Sample filters and separate host branches

Stage	Listings	Hosts	Role
Raw listings	25,728	14,113	Source snapshot
After scope filters	6,891	3,488	Before host partition
Analysis before support	5,549	2,809	Before 50-listing rule
Final analysis	3,873	1,726	14 supported segments
Reference before support	1,342	679	Separate benchmark hosts
Final reference	906	426	Same 14 segments; defines P75

Methodology and Analytical Approach

A fixed hash of host identifiers reserves approximately 20% of hosts for benchmark development. These hosts enter no model training, evaluation or sensitivity analysis. After removing them, analysis segments must contain at least 50 listings. Eligible reference listings in those supported segments supply one pooled P75, using linear interpolation and upward integer rounding. That cutoff remains fixed across segments and sensitivity samples.

Descriptive tables report listing and distinct-host counts, review quantiles, observed attainment and attribute profiles. Pointwise 90% intervals resample hosts to account for properties sharing an operator. Wider dwelling definitions and unrestricted review histories assess how the chosen scope affects comparisons.

Baseline logistic regression and random forest use LGA, dwelling–bedroom configuration, capacity, bathrooms and amenity counts. Six fixed extensions add coordinates, beds and nine facilities and also test histogram gradient boosting. All comparisons use the same 3,873 listings and five test folds that keep each host in one fold. Each fold learns missing-value treatment and encoding only from its training data. Ratings, review-derived predictors and host badges are excluded; price and minimum stay appear only in labelled operating-controls variants.

Model assessment uses four measures. AUC measures ranking ability; average precision measures how accurately the model identifies listings reaching the benchmark; the Brier score measures probability error, where lower is better; and calibration checks whether predicted probabilities agree with observed rates. Every model is also compared with two references on the same folds: a training-prevalence forecast, and a segment-rate baseline that gives each

validation listing its segment's attainment rate among training hosts, shrunk towards the training rate by a prior weight of ten fixed before the run. Two boosting variants adjust probabilities using three host-separated groups within each training fold, without using held-out outcomes. R supports cleaning and descriptive analysis; Python and scikit-learn support modelling.

Analysis Plan and Progress to Date

The complete raw-data workflow and independent checks now agree on review counts, samples, benchmark and baseline metrics. Baseline logistic AUC is 0.573 and forest AUC is 0.560. Detailed property features improve forest AUC to 0.640, average precision to 0.350 and Brier score to 0.181, versus baseline logistic Brier 0.189. Its ten-bin calibration error is 1.63 percentage points. Boosting achieves AUC 0.647 but Brier 0.182. We provisionally prefer the enhanced forest for probability ranking because it has the lowest Brier score among the property-only extensions; discrimination remains moderate.

The enhanced forest ranks Melbourne three-bedroom apartments first: mean probability 34.4%, observed attainment 36.6%, and 306 listings. Resampling hosts gives a conditional 95% interval of 33.0–35.8%; this interval holds the fitted model and review cutoff fixed. Removing the history restriction gives 6,675 analysis listings; removing the price filter gives 5,720. Adding current price and minimum stay to enhanced boosting raises AUC to 0.749. These contemporaneous settings provide useful context but do not establish a causal effect or a pre-opening forecast. Probability adjustment does not improve every metric.

The segment-rate baseline reaches AUC 0.581 and Brier score 0.187. Both original property-only models fall below it, exceeding its AUC in one of five folds; the enhanced forest exceeds it in every fold, by 0.059 AUC overall. Every model places the same three segments first, so the models add listing-level discrimination rather than a different segment order. Across 20 repeated host partitions the enhanced forest exceeds the baseline in all 20 (AUC 0.589–0.648); Melbourne three-bedroom apartments rank first in every partition and the same top three recur in 19.

All six extensions and the baseline are reported. Their comparison follows baseline inspection, so model preference is exploratory and there is no untouched final test. Next steps are to choose models within the training data and recalculate ranking intervals while refitting the model. Findings describe existing listings' preceding-year review activity, not future income or profitability. The README reproduces this report and links the code, validation and outputs.

References

Inside Airbnb. (2026). *Melbourne listings, calendar and reviews* [June 2026 dataset supplied through CMCE30005 LMS].

Scikit-learn developers. (n.d.). *Cross-validation and Probability calibration*. Retrieved 13 September 2026. [Grouped validation](#); [Calibration guidance](#).